



CHANGEMAKERS

São Paulo, Brazil Soccer Infrared - ROBOCUP JR.

1.0 ABSTRACT

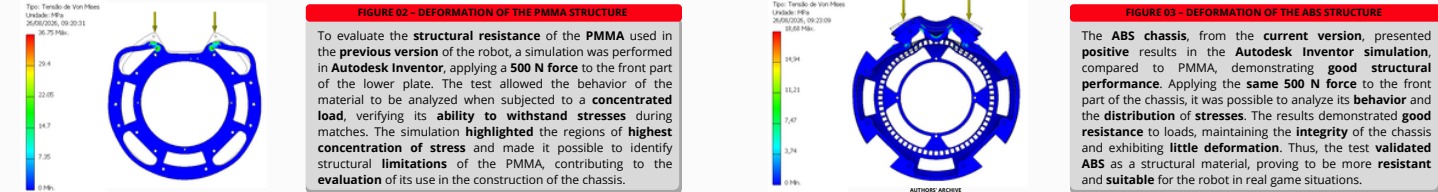
We are the SESI ChangeMakers team from SESI São Paulo, Brazil, and we are currently in our second season competing in RoboCup Junior Soccer Infrared. We have been continuously working on the design of our current robots for approximately one year: we began the planning process in November 2025, focusing on improvements in hardware, software, mechanics, and system integration, with the goal of increasing reliability, efficiency, and performance during matches, as well as developing our own solutions that are replicable and can be continuously improved. Regarding the mechanical design, the chassis was entirely designed by the team and manufactured from ABS using 3D printing, aiming to achieve a balance between structural strength, weight reduction, and dynamic performance. The design prioritizes modularity: it is composed of four modules, facilitating maintenance, component replacement, and the implementation of future improvements. The control system, developed in C++, was structured in a modular and flexible manner, facilitating development, maintenance, and integration among the different onboard modules. The electronic architecture was also improved: we designed our own Printed Circuit Boards (PCBs), which are one of the main distinguishing features of the project. The upper, main, kicker, and lower board were developed providing more efficient component organization, greater reliability in electrical connections, and better integration among the robot's systems. Information regarding team contact details, files, electrical schematics, and CAD models is available through the links in the footer of the banner.

2.0 HARDWARE

ENGINEERING AND MANUFACTURING DESIGN (MECHANICAL AND ELECTRONICS)

2.1 MECHANICAL

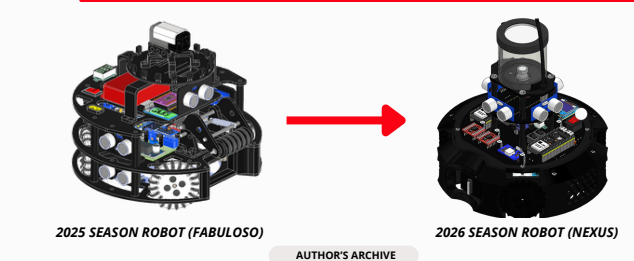
In the previous season, the robot's chassis was constructed from PMMA (acrylic). For the current season, we opted to use ABS, aiming to achieve a structure with lower mass and greater mechanical strength against the loads experienced during matches. The choice of ABS was evaluated using the stress simulation environment in Autodesk Inventor, allowing us to compare the structural behavior of both materials under equivalent conditions. For the analysis, a 500 N force was applied to the robots' plate/chassis, perpendicular to its surface, while maintaining the same loading direction, position, and magnitude for both the PMMA and ABS models, simulating the impacts that occur during matches. The comparison allowed us to numerically and descriptively evaluate parameters such as stress, strain, and displacement, making it possible to identify differences in the structural behavior of the materials. Therefore, the choice of ABS was not based solely on its theoretical characteristics of strength and low mass, but also on a comparative analysis performed using the actual geometry implemented in the robot. The simulation results were used as part of the material selection validation process, relating the resulting structural performance to the need for reduced mass and resistance to mechanical loads during robot operation.



2.2 ELETRONICAL

One of the main evolutions of the project was the replacement of acrylic sheets with PCBs (Printed Circuit Boards) as structural elements, integrated into the chassis architecture. This improved the organization of the modules and promoted a reduction in wiring, resulting in a more compact, robust system that is less susceptible to failures. In addition, there was a reduction in mass, better distribution of mechanical loads, and improvements in assembly, maintainability, and modularity.

FIGURE 04 - COMPARISON BETWEEN ROBOTS FROM DIFFERENT SEASONS



2.3 KICKER BOARD

The kicker board was designed for the controlled storage and release of energy (Figure 09), enabling the efficient execution of the ball kick. The system is powered by 12 V from the main battery and uses an XL6009 boost converter to raise the voltage to an appropriate level for charging the power circuit. The energy is stored in a bank of two 2200 µF capacitors, capable of delivering high currents over a short period of time. The trigger command operates at 3.3 V and is electrically isolated by a PC817 optocoupler, protecting the control electronics against noise and voltage spikes. The control signal is amplified by a transistor stage, responsible for the rapid activation of the IRL244N MOSFET, which acts as an electronic switch for the instantaneous discharge of the capacitors into the solenoid. The high current generates the magnetic field required to propel the plunger and perform the kick. After firing, the system automatically recharges, ensuring readiness for subsequent actions.

2.4 LOWER BOARD

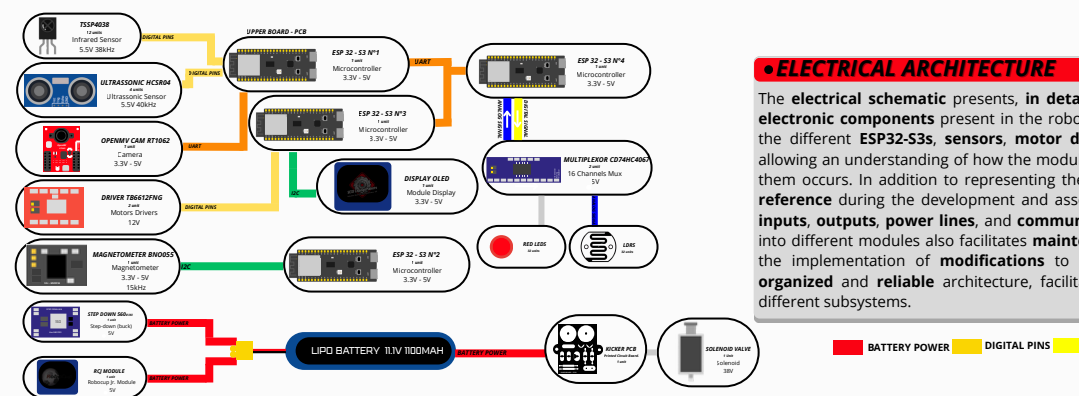
The lower board was developed for the field line detection system (Figure 10). The assembly uses 32 LDR sensors combined with 32 red LEDs, responsible for the controlled illumination of the surface and the stability of the measurements. Each sensor and each light emitter has a dedicated resistor, totaling 64 resistors, ensuring the correct electrical operation of the circuit. Signal processing is performed by an ESP32-S3 microcontroller, responsible for the acquisition and processing of information. The readings are obtained through two CD74HC4067 multiplexers, allowing for expanded inputs and system organization. The identification of the markings is based on the difference in reflectance between the white of the lines and the green of the field, with the use of red LEDs to increase contrast. The processed information is then transmitted to the second ESP32-S3, located on the main board, where it is integrated into the overall control logic of the robot.

2.5 MAIN BOARD

The main board was designed to serve as the control and integration core of the robot's electronic system (Figure 08), concentrating the main processing, communication, and actuation modules, which ensures efficient interconnection between the different subsystems. The circuit features dedicated connectors for the kicker board, upper module, and lower module, enabling organized communication while also facilitating maintenance and future system expansions. For actuator control, two TB6612FNG motor drivers were integrated, responsible for the precise and efficient control of the drive motors. The robot's orientation is managed through a BNO055 compass sensor, which provides directional information and contributes to navigation and stabilization on the field. Communication with the official competition systems is ensured through the integration of the RC (RoboCup Junior) module. In addition, the system incorporates an OLED display, used to demonstrate diagnostic information, operational states, and robot adjustments. Processing is carried out by two ESP32-S3 microcontrollers (the actuator and the processor), which operate in a distributed manner, optimizing performance and communication between the electronic modules. Power is supplied through a battery connector, with the voltage regulated by a Mini 560 Step-Down converter, ensuring a stable, efficient, and suitable power supply for all system components.

2.6 UPPER BOARD

For this season, an upper board was developed to integrate the system's main sensors (Figure 07). Twelve TSSP4038 sensors were mounted on the underside of the plate, with the function of detecting the infrared signals emitted by the ball. On the same plate, on the upper side, four HC-SR04 ultrasonic sensors were integrated and arranged in a cross configuration, with the purpose of locating the robot on the field by identifying walls and the goal through distance measurements, along with an ESP32-S3 for processing. In addition, an OpenMV Cam RT1062 camera was integrated, operating together with a convex mirror, enabling omnidirectional vision of the environment. This system allows the identification of the field and the ball through image processing, contributing to the improvement of game strategies.



TOTAL ROBOT COST: \$935 (American Dollar)
The stated amount corresponds to the manufacturing cost of one complete robot.

4.0 SOFTWARE
4.1 ROBOT STRATEGY ORGANIZED INTO FLOWCHARTS

